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OPERATION PEDAL DEVICE FOR VEHICLE

Abstract

When a dash panel P is displaced toward the rear of a vehicle upon vehicle collision, a small collision lever **60** is displaced to a release position at which a load corresponding to contact of a collision bracket **200** with a second contact portion **60A** is applied to a swaging pin **15**. When fixation between a rotation lever **50** and a second arm **40** via the small collision lever **60** by the swaging pin **15** is released, a step portion **21** is displaced to the vehicle front side.

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to a vehicular operation pedal device, which suppresses a step portion of an operation pedal mechanism from moving back to a vehicle rear side when a vehicle component is displaced to the vehicle rear side due to a vehicle collision (hereinafter, it is referred to as “prevention of backward movement of the step portion of the operation pedal mechanism upon vehicle collision”).

BACKGROUND ART

[0002] Patent Literature 1 describes a vehicular operation pedal device, which realizes prevention of backward movement of a step portion. The vehicular operation pedal device includes a support member, the step portion, an operation pedal, a rotation lever holding an operating rod, and a fixing member. The rotation lever is supported by a rotation shaft member so as to be rotatable relative to the operation pedal, and is fixed to the operation pedal by the fixing member. Upon vehicle collision, a first vehicle component to which the support member is fixed moves back to a vehicle rear side, and the rotation lever contacts a second vehicle component. Accordingly, a predetermined load acts on the fixing member via the rotation lever, and the fixing member releases the fixation between the rotation lever and the operation pedal. With this configuration, the rotation lever rotates to the vehicle front side, preventing the step portion from moving back to the vehicle rear side.

CITATION LIST

Patent Literature

[0003] Patent Literature 1: JP-A-2020-095373

SUMMARY OF INVENTION

Problems to be Solved by Invention

[0004] In the vehicular operation pedal device described in Patent Literature 1, it is necessary to make a load for releasing the fixation between the rotation lever and the operation pedal by the fixing member greater than a load applied to the fixing member when stepping operation is performed on the step portion. For this reason, it is necessary to design, e.g., the rotation shaft position and shape of each lever in consideration of two loads which are a load applied to the fixing member via the rotation lever by the stepping operation relative to the step portion and a load applied to the fixing member via the rotation lever upon vehicle collision. Thus, there is a concern that the degree of flexibility in design for a vehicular operation pedal device is lowered.

[0005] The present invention has been made in view of the above-described problems, and it is an object of the present invention to provide a vehicular operation pedal device, which is configured so that the degree of flexibility in design can be enhanced in the vehicular operation pedal device that achieves prevention of backward movement of a step portion of an operation pedal mechanism upon vehicle collision.

Solution to Problems

[0006] A vehicular operation pedal device as disclosed in the present specification to solve the above-mentioned problem includes a support member fixed to a first vehicle component, an operation pedal mechanism provided on the support member and having a step portion that is rotatable relative to the support member and an operation lever that rotates toward a vehicle front side when the step portion is stepped on toward the vehicle front side, a rotation lever having a first contact portion extending to a vehicle rear side, supported on a vehicle lower side with respect to the first contact portion by a first rotation shaft portion so as to be rotatable toward the vehicle front side relative to the operation lever, and holding an operating rod protruding to the vehicle rear side from the first vehicle component, a differential lever having a second contact portion extending to the vehicle rear side and supported on the vehicle lower side with respect to the second contact portion so as to be rotatable toward the vehicle front side relative to the rotation lever by a second rotation shaft portion, and a fixing member that fixes the differential lever and the operation lever.

The differential lever is, when fixed to the operation lever by the fixing member, at a restriction position at which the second contact portion protrudes to the vehicle rear side beyond the first contact portion of the rotation lever. When the first vehicle component is displaced toward the vehicle rear side upon vehicle collision, a second vehicle component positioned on the vehicle rear side with respect to the first vehicle component contacts the second contact portion, so that the second contact portion is positioned on the vehicle front side with respect to the first contact portion, and the differential lever is displaced to a release position at which a load corresponding to contact of the second vehicle component with the second contact portion is applied to the fixing member, and fixation between the differential lever and the operation lever by the fixing member is released, so that the second vehicle component contacts the first contact portion and the rotation lever rotates toward the vehicle front side at the first rotation shaft portion.

Effects of Invention

[0007] According to the present invention, it is possible to provide a vehicular operation pedal device, which has a high degree of flexibility in design in the vehicular operation pedal device that achieves prevention of backward movement of the step portion.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0008] FIG. 1 is a side view illustrating an outline configuration of a brake pedal device according to a first embodiment;

[0009] FIG. 2 is an exploded view of part of the brake pedal device;

[0010] FIG. 3 is a view illustrating a section of the brake pedal device taken along a line A-A in FIG. 1;

[0011] FIG. 4 is a side view illustrating the outline configuration of the brake pedal device;

[0012] FIG. 5 is a side view illustrating the outline configuration of the brake pedal device;

[0013] FIG. 6 is a side view illustrating the outline configuration of the brake pedal device;

[0014] FIG. 7 is an exploded view of part of a brake pedal device according to a second embodiment; and

[0015] FIG. 8 is an exploded view of part of a brake pedal device according to a modification of the second embodiment.

DESCRIPTION OF EMBODIMENTS

[0016] Hereinafter, a vehicular operation pedal device will be described by taking a brake pedal device for a common use brake as an example. In each drawing used in the following description, part of a basic configuration is omitted, and the dimensional ratios, etc. of the illustrated elements are not necessarily accurate.

[0017] In the following description, the brake pedal device will be described using a direction in a state where the brake pedal device is mounted on a vehicle. Specifically, the front of the vehicle will be described as “vehicle front,” the rear will be described as “vehicle rear,” the upper side of the vehicle will be described as “vehicle upper,” and the lower side of the vehicle will be described as “vehicle lower.” The left-and-right direction of the vehicle will be referred to as a “vehicle width direction.” In each figure, the “vehicle front” is simply referred to as “front,” the “vehicle rear” is simply referred to as “rear,” the “vehicle upper” is simply referred to as “upper,” and the “vehicle lower” is simply referred to as “lower.” One side in the “vehicle width direction” will be described as “left,” and the other side will be described as “right.”

(1-1) Outline of First Embodiment

[0018] As illustrated in FIGS. 1 and 2, a brake pedal device 1 mainly includes a pedal bracket 10, an operation pedal mechanism 20, a rotation lever 50, a small collision lever 60, and a pedal force detection device 80.

[0019] The pedal bracket **10** is fixed to a dash panel P. The dash panel P forms part of the vehicle and is positioned on the vehicle front side with respect to the operation pedal mechanism **20**. The pedal bracket **10** includes a pair of side plates **16**. The pair of side plates **16** faces each other at a predetermined interval in the vehicle width direction and is fixed to the dash panel P with bolts, or the like.

[0020] In FIGS. **1** and **2**, among the pair of side plates **16**, only the side plate **16** on the left side in the vehicle width direction is illustrated, and the side plate on the right side in the vehicle width direction is not illustrated. The same also applies to FIGS. **4** to **6** described later.

[0021] An operating rod **90** protrudes from the dash panel P to the vehicle rear side. An end portion of the operating rod **90** positioned on the vehicle front side with respect to the dash panel P is coupled to the operation pedal mechanism **20** with a clevis **70** and a clevis pin **71**. The operating rod **90** is displaced toward the vehicle front side in association with stepping operation on a step portion **21** of the operation pedal mechanism **20**, so that the operation force at the time of the stepping operation is transmitted to a braking device or a control device that controls the driving state of the vehicle via a hydraulic circuit, an electronic circuit, or the like.

[0022] The operation pedal mechanism **20** is arranged between the pair of side plates **16** of the pedal bracket **10**. The operation pedal mechanism **20** is a link type operation pedal mechanism and mainly has the step portion **21**, a first arm **22**, a link member **30**, a second arm **40**, an operation shaft portion **11**, and an intermediate shaft portion **12**.

[0023] The first arm **22** is made of metal and is rotatably supported relative to the pedal bracket **10** at an upper end portion **22A** in a longitudinal direction which is a direction in which the first arm **22** extends by the operation shaft portion **11** penetrating the first arm **22** in the vehicle width direction. The step portion **21** is provided at a lower end portion **22B** of the first arm **22** in the longitudinal direction. With this configuration, when a driver of the vehicle steps on the step portion **21** toward the vehicle front side (hereinafter referred to as “stepping operation”), the first arm **22** is allowed to rotate about the operation shaft portion **11** toward the vehicle front side. Note that in FIG. **1**, the step portion **21** is at an initial position in a state of the stepping operation being not performed.

[0024] As illustrated in FIG. **2**, the second arm **40** is a thin metal plate-shaped member and has an offset portion **40A**, an intermediate portion **40B**, and a lower end portion **40C** formed continuously in a longitudinal direction which is a direction in which the second arm **40** extends. The offset portion **40A** is a portion extending from the intermediate portion **40B** in the longitudinal direction of the second arm **40** in such a manner as to be offset to the vehicle rear side. With this configuration, the vehicle upper side of the second arm **40** is in an inverted L-shape formed by the offset portion **40A** and the intermediate portion **40B**.

[0025] As illustrated in FIG. **2**, a boss **41** through which the intermediate shaft portion **12** penetrates is formed at the lower end portion **40C** of the second arm **40**. The intermediate shaft portion **12** is a shaft portion rotatably supporting the second arm **40** relative to the pedal bracket **10**. In the lower end portion **40C**, a through-hole **42** through which a link pin **31** of the link member **30** penetrates is formed on the intermediate portion **40B** side (i.e., vehicle upper side) with respect to the boss **41**. Note that the intermediate shaft portion **12** is arranged between the pair of side plates **16** in a state of being substantially horizontal and substantially parallel with the vehicle width direction with respect to the operation shaft portion **11**.

[0026] A clearance hole **43** which is an opening penetrating the second arm **40** in the vehicle width direction is formed in the intermediate portion **40B**. In the intermediate portion **40B**, a bearing hole **44** through which a first rotation shaft portion **13** penetrates is formed on the offset portion **40A** side (i.e., vehicle upper side) with respect to the clearance hole **43**. A through-hole **45** through which a swaging pin **15** described later penetrates is formed in the offset portion **40A**.

[0027] The link member **30** is a member that couples the first arm **22** and the second arm **40** to each other. The link member **30** couples the first arm **22** and the second arm **40** to each other in

such a manner that the link pin **31** penetrates through the first arm **22**, and the link pin **32** penetrates through the through-hole **42** of the second arm **40**. With this configuration, the operation pedal mechanism **20** is configured to transmit rotation of the first arm **22** at the operation shaft portion **11** by the stepping operation on the step portion **21** through the link member **30** so that the second arm **40** can rotate at the intermediate shaft portion **12**.

[0028] The rotation lever **50** is rotatably attached to the second arm **40**. As illustrated in FIG. 2, the rotation lever **50** is a thin metal plate-shaped member and has a lower end portion **50C**, an intermediate portion **50B**, and a first contact portion **50A** formed continuously in a longitudinal direction in which the rotation lever **50** extends. The intermediate portion **50B** is a portion having an arrangement region on which the pedal force detection device **80** can be arranged. The arrangement region is a region not overlapping with the second arm **40** when the rotation lever **50** is arranged on the second arm **40**, in other words, a region of the intermediate portion **50B** of the rotation lever **50** not overlapping with the offset portion **40A**. The first contact portion **50A** is a portion bent and extending from the intermediate portion **50B** to the vehicle rear side in the longitudinal direction of the rotation lever **50**.

[0029] As illustrated in FIG. 2, a clevis holding member **51** on which the clevis **70** is held is fixed to the lower end portion **50C**. Details of the clevis holding member **51** will be described later. In the lower end portion **50C**, a bearing hole **52** through which the first rotation shaft portion **13** penetrates is formed on the intermediate portion **50B** side (i.e., vehicle upper side) with respect to the clevis holding member **51** in the longitudinal direction of the rotation lever **50**. The first rotation shaft portion **13** is a shaft that rotatably supports the rotation lever **50** relative to the second arm **40**. A bearing hole **53** through which a second rotation shaft portion **14** penetrates is formed in the intermediate portion **50B**. The second rotation shaft portion **14** is a shaft that rotatably supports the small collision lever **60** relative to the rotation lever **50**. Specifically, in the rotation lever **50**, the bearing hole **53** for the second rotation shaft portion **14** is formed on the vehicle upper side with respect to the arrangement region where the pedal force detection device **80** is arranged. In other words, the arrangement region is formed between the bearing hole **52** for the first rotation shaft portion **13** and the bearing hole **53** for the second rotation shaft portion **14**.

[0030] The small collision lever **60** is rotatably supported on the rotation lever **50**. The small collision lever **60** is a thin metal plate-shaped member and has a lower end portion **60C**, an intermediate portion **60B**, and a second contact portion **60A** formed continuously in a longitudinal direction in which the small collision lever **60** extends. The second contact portion **60A** is a portion bent and extending from the intermediate portion **60B** to the vehicle rear side.

[0031] A through-hole **61** through which the swaging pin **15** penetrates is formed in the lower end portion **60C** of the small collision lever **60**. The swaging pin **15** is a shaft for fixing the small collision lever **60** to the second arm **40** which is a component of the operation pedal mechanism **20**. A through-hole **62** through which the second rotation shaft portion **14** penetrates is formed in the intermediate portion **60B**. In the present embodiment, the swaging pin **15** is swaged so as not to fall off from the second arm **40** and the small collision lever **60**. In addition, the material and diameter of the swaging pin **15** are designed such that the swaging pin **15** is broken when a predetermined load is applied to the swaging pin **15**.

[0032] In the above-described configuration, the rotation lever **50** is rotatably supported relative to the second arm **40** by the first rotation shaft portion **13**. The small collision lever **60** is rotatably supported relative to the rotation lever **50** by the second rotation shaft portion **14**. The swaging pin **15** penetrates through the second arm **40** and the small collision lever **60** to restrict rotation of the small collision lever **60** at the second rotation shaft portion **14**. With this configuration, the rotation lever **50** is fixed to the second arm **40** via the small collision lever **60** whose rotation at the second rotation shaft portion **14** is restricted by the swaging pin **15**. As a result, the rotation lever **50** can be rotated in the same direction as the rotation direction of the second arm **40** in association with rotation of the second arm **40** at the intermediate shaft portion **12**.

[0033] FIG. 3 illustrates a section of the brake pedal device 1 taken along a line A-A in FIG. 1. The clevis 70 is fixed to the tip end of the operating rod 90. The clevis 70 is swingably held via the clevis pin 71 by the clevis holding member 51 housed in the clearance hole 43 of the second arm 40. Specifically, the clevis holding member 51 has a through-hole 51a through which the clevis pin 71 penetrates. The inner diameter of the through-hole 51a of the clevis holding member 51 is greater than the diameter of the clevis pin 71. The clevis 70 has a pair of side plate portions 72, and the clevis pin 71 penetrates through the clevis 70 together with the clevis holding member 51 in a state where the second arm 40 and the clevis holding member 51 positioned in the clearance hole 43 of the second arm 40 are sandwiched by the pair of side plate portions 72. Note that a tip end portion of the clevis pin 71 penetrating through the side plate portions 72 is configured not to fall off from the clevis holding member 51 and the clevis 70 by a clip, or the like. With this configuration, when a reactive force is applied from the operating rod 90 to the clevis 70, the clevis 70 can swing in a direction in which the operating rod extends (i.e., the direction of the reactive force) by an amount corresponding to a clearance between the through-hole 51a of the clevis holding member 51 and the clevis pin 71.

[0034] As illustrated in FIGS. 1 and 2, the pedal force detection device 80 is attached to the intermediate portion 50B of the rotation lever 50, and when the stepping operation is performed on the step portion 21, the rotation lever 50 and the pedal force detection device 80 cooperate with each other such that a pedal force based on the stepping operation is detected.

[0035] As illustrated in FIG. 1, the pedal force detection device 80 has a pedal force SW 81 and a swing lever 82. The pedal force SW 81 is a known circuit and outputs a signal corresponding to the pedal force according to the amount of pressing of a detection shaft. The pedal force SW 81 is fixed to the intermediate portion 50B such that the detection shaft protrudes further forward of the vehicle front side than the rotation lever 50. The swing lever 82 has a receiving portion 83 facing the detection shaft of the pedal force SW 81 and is swingably held by the clevis holding member 51 by being fixed to the clevis pin 71. The receiving portion 83 of the swing lever 82 is arranged so as to face the detection shaft of the pedal force SW 81 via a spring, swings in accordance with the reactive force from the operating rod 90, and contacts the detection shaft of the pedal force SW 81. In the present embodiment, the swing lever 82 is fixed to the clevis pin 71 on the side opposite to the second arm 40 in the vehicle width direction.

(1-2) Operation of First Embodiment During Stepping Operation

[0036] When the stepping operation is performed on the step portion 21, the first arm 22 rotates about the operation shaft portion 11. At this time, the first arm 22 rotates in a predetermined direction (clockwise direction in FIG. 1) about the operation shaft portion 11, and therefore, the rotation of the first arm 22 is transmitted to the second arm 40 via the link member 30. The second arm 40 rotates in a predetermined direction (counterclockwise direction in FIG. 1) about the intermediate shaft portion 12 in association with rotation of the first arm 22. Thus, the rotation lever 50 whose rotation at the first rotation shaft portion 13 relative to the second arm 40 is restricted by the small collision lever 60 rotates in the same direction as the rotation direction of the second arm 40, and displaces the tip end of the operating rod 90 connected to the clevis 70 to the vehicle front side.

[0037] A reactive force FA from the operating rod 90 to the vehicle rear side in the direction in which the operating rod 90 extends acts on the clevis pin 71 via the clevis 70. As described above, the inner diameter of the through-hole 51a of the clevis holding member 51 is greater than the diameter of the clevis pin 71, and the clevis 70 and the swing lever 82 swing by the reactive force from the operating rod 90. With this configuration, the receiving portion 83 of the swing lever 82 contacts the detection shaft of the pedal force SW 81, and the pedal force SW 81 can detect a pedal force corresponding to the amount of push-in of the detection shaft. The pedal force SW 81 outputs a signal corresponding to the detected pedal force to a not-illustrated controller.

[0038] At this time, a load corresponding to the reactive force from the operating rod 90 acts on the

swaging pin **15**. For example, when the reactive force F_A from the operating rod **90** acts on the clevis pin **71** by the step portion **21** being stepped on to the vehicle front side to the maximum extent by a driver of the vehicle, a first load F_1 acting on the swaging pin **15** is expressed by Expression (I) below.

$$[00001] F_1 = F_A \times \sin\theta_1 \times LA / LB \quad \text{Expression(I)}$$

[0039] Here, θ_1 is an angle at which a straight line connecting the clevis pin **71** and the swaging pin **15** intersects the direction of the reactive force F_A from the operating rod **90** at the clevis pin **71**. LA is a distance from the clevis pin **71** to the first rotation shaft portion **13**. LB is a distance from the first rotation shaft portion **13** to the swaging pin **15**.

[0040] In a case where the driver steps on the step portion **21** toward the vehicle front side to the maximum extent, and the driver further steps on the step portion **21** toward the vehicle front side, an operation load acting on the step portion **21** increases, the reactive force F_A also increases, and the first load F_1 also increases. Thus, in the present embodiment, the load acting on the swaging pin **15** in a case where the operation load is the maximum in design is the first load F_1 .

(1-3) Operation of First Embodiment Upon Vehicle Collision

[0041] As illustrated in FIG. 4, when the dash panel **P** is displaced to the vehicle rear side upon vehicle collision, a collision bracket **200** of an instrument panel reinforcement contacts the second contact portion **60A** of the small collision lever **60**. Accordingly, a load acts on the swaging pin **15**. For example, in a case where a collision force F_B acts on the small collision lever **60** at a contact point **C** at which the collision bracket **200** and the small collision lever **60** contact with each other, a second load F_2 acting on the swaging pin **15** is expressed by Expression (II) below.

$$[00002] F_2 = F_B \times \cos\theta_2 \times LC / LD \quad \text{Expression(II)}$$

[0042] Here, θ_2 is an angle at which a straight perpendicular line connecting the contact point **C** and the swaging pin **15** to each other intersects the direction of the collision force F_B at the contact point **C** of the small collision lever **60**. LC is a distance from the second rotation shaft portion **14** to the contact point **C**. LD is a distance from the swaging pin **15** to the second rotation shaft portion **14**.

[0043] A relationship between the first load F_1 and the second load F_2 is expressed by Expression (III) below.

$$[00003] F_2 > F_1 \quad \text{Expression(III)}$$

[0044] Expression (III) indicates that by designing the respective distances LA , LB , LC , and LD such that the second load F_2 is greater than the first load F_1 , it is possible to prevent release of the fixation between the rotation lever **50** and the second arm **40** by the swaging pin **15** during operation of the brake pedal device **1**. In the present embodiment, the swaging pin **15** is designed such that a load when the swaging pin **15** is broken is the second load F_2 or more.

[0045] As illustrated in FIG. 4, distances for calculating the first load F_1 are the distances LA and LB between the shaft portions in the rotation lever **50**, whereas the distances LC and LD for calculating the second load F_2 are a distance between the contact point **C** and the shaft portion in the small collision lever **60** and a distance between the shaft portions in the small collision lever **60**. This is because the second load F_2 does not directly act on the rotation lever **50** upon vehicle collision. The shapes of the second arm **40** and the rotation lever **50** and the positions of the shaft portions can be designed within a range satisfying Expressions (I) to (III) described above.

[0046] When the second load F_2 exceeds a reference load greater than the first load F_1 , the swaging pin **15** is cut off by the second load F_2 , the small collision lever **60** becomes rotatable at the second rotation shaft portion **14**, and the fixation between the rotation lever **50** and the second arm **40** via the small collision lever **60** is released, as illustrated in FIG. 5. Thereafter, continuance of the contact of the collision bracket **200** with the second contact portion **60A** of the small collision lever **60** allows the second contact portion **60A** of the small collision lever **60** to rotate to the vehicle front side at the second rotation shaft portion **14** and allows the small collision lever **60**

to move from a restriction position to a release position. The restriction position is a position at which in the small collision lever **60**, the second contact portion **60A** protrudes to the vehicle rear side beyond the first contact portion **50A**. The release position is a position at which in the small collision lever **60**, the second contact portion **60A** is positioned on the vehicle front side with respect to the first contact portion **50A**. With this configuration, the first contact portion **50A** of the rotation lever **50** contacts the collision bracket **200**.

[0047] As illustrated in FIG. **6**, when the collision bracket **200** contacts the first contact portion **50A**, the rotation lever **50** rotates about the first rotation shaft portion **13** to the vehicle front side (counterclockwise direction in FIG. **6**).

[0048] In association with the rotation lever **50**, the tip end of the operating rod **90** is displaced toward the vehicle front side via the clevis **70**. Since the second arm **40** rotates about the intermediate shaft portion **12** toward the vehicle front side (counterclockwise direction in FIG. **6**) in response to rotation of the rotation lever **50**, the step portion **21** of the first arm **22** is displaced to the vehicle front side. Note that the step portion **21** indicated by a chain double-dashed line indicates an initial position of the step portion **21** when the stepping operation is not performed.

[0049] Further, along with rotation of the rotation lever **50** at the intermediate shaft portion **12**, the clevis holding member **51** housed in the clearance hole **43** rotates about the first rotation shaft portion **13**, inside the clearance hole **43**. In the present embodiment, the first rotation shaft portion **13** is positioned on the vehicle upper side with respect to a line segment in a direction in which the operating rod **90** extends when the small collision lever **60** is at the restriction position and the step portion **21** is at an initial position at which the stepping operation is not performed on the step portion **21**. In other words, the first rotation shaft portion **13** is positioned on the vehicle upper side with respect to the clearance hole **43** of the second arm **40** in which the clevis holding member **51** is housed.

[0050] It is assumed that in the second arm **40**, the first rotation shaft portion **13** is on the vehicle rear side with respect to the clearance hole **43**. In this case, the movement locus of the clevis holding member **51** in association with rotation of the rotation lever **50** upon vehicle collision shifts in an arc shape toward the vehicle lower side centered on the first rotation shaft portion **13**. Thus, the second arm **40** needs to be formed with the arc-shaped clearance hole **43** facing the vehicle lower side according to the movement locus of the clevis holding member **51**. However, in the second arm **40**, the dimension of the intermediate portion **40B** on the vehicle lower side increases in order to ensure the thickness around the clearance hole **43**, and for example, it may be necessary to shift the position of the first arm **22** through which the operation shaft portion **11** penetrates, to the vehicle lower side. In contrast, in the present embodiment, in the second arm **40**, the first rotation shaft portion **13** is positioned on the vehicle upper side with respect to the clearance hole **43**. Thus, the movement locus of the clevis holding member **51** shifts to the vehicle rear side centered on the first rotation shaft portion **13** in accordance with rotation of the rotation lever **50**. Thus, the clearance hole **43** facing the vehicle rear side according to the movement locus of the clevis holding member **51** is formed in the second arm **40**. With this configuration, it is easy to ensure the thickness around the clearance hole **43** without increasing the dimensions of the intermediate portion **40B** of the second arm **40** toward the vehicle lower side. Thus, the position of the second arm **40** and the position of the operation shaft portion **11** of the first arm **22** are less likely to interfere with each other, and therefore, an increase in the dimension of the brake pedal device **1** can be suppressed.

[0051] In the present embodiment, the second arm **40** is an example of an operation lever. The small collision lever **60** is an example of a differential lever. The swaging pin **15** is an example of a fixing member. The clevis **70** is an example of a coupling member. The clevis holding member **51** is an example of a holding member. The pedal bracket **10** is an example of a support member. The dash panel **P** is an example of a first vehicle component of a vehicle, and the collision bracket **200** is an example of a second vehicle component.

(1-4) Summary of First Embodiment

[0052] In the brake pedal device **1** described above, when the collision bracket **200** contacts the second contact portion **60A** of the small collision lever **60** and the second contact portion **60A** is displaced toward the release position, accordingly, the swaging pin **15** is broken due to a load applied to the swaging pin **15**, and fixation of the rotation lever **50** to the second arm **40** via the small collision lever **60** is released. At this time, the small collision lever **60** contacts the collision bracket **200**, and thus the second load **F2** upon vehicle collision does not directly act on the rotation lever **50**. Thereafter, when the collision bracket **200** contacts the first contact portion **50A** of the rotation lever **50**, the rotation lever **50** rotates to the vehicle front side at the first rotation shaft portion **13**, and the step portion **21** of the first arm **22** rotates to the vehicle front side accordingly. Thus, the distances **LA**, **LB** for calculating the first load **F1** acting on the swaging pin **15** upon operation of the step portion **21** and the distances **LC**, **LD** for calculating the second load **F2** acting on the swaging pin **15** upon vehicle collision can be determined independently. As a result, the degree of flexibility in design can be increased for the shapes of the second arm **40** and the rotation lever **50** and the positions of the shaft portions.

[0053] The clevis holding member **51** holds the clevis **70** on the rotation lever **50** in a state of being housed in the clearance hole **43** of the second arm **40**. The first rotation shaft portion **13** is positioned on the vehicle upper side with respect to the line segment in the direction in which the operating rod **90** extends when the small collision lever **60** is at the restriction position and the step portion **21** is at the initial position at which the stepping operation is not performed on the step portion **21**. With this configuration, it is easy to ensure the thickness around the clearance hole **43** in the second arm **40**. The shape of the second arm **40** can be designed without increasing the dimensions of the second arm **40**, and therefore, an increase in the dimensions of the brake pedal device **1** can be suppressed.

[0054] The clevis holding member **51** holds the clevis **70** on the intermediate portion of the rotation lever such that the clevis **70** can swing in a direction in which the reactive force from the operating rod **90** is applied, and holds the swing lever **82** of the pedal force detection device **80** such that the swing lever **82** swings together with the clevis **70**. With this configuration, also in the brake pedal device **1** including the pedal force detection device **80**, the degree of flexibility in design can be increased for the shapes of the second arm **40** and the rotation lever **50** and the positions of the shaft portions.

[0055] The swaging pin **15** penetrates through the offset portion **40A** of the second arm **40**. With this configuration, in a state of the rotation lever **50** being fixed to the second arm **40** via the small collision lever **60**, a region not overlapping with the second arm **40** can be formed in the rotation lever **50**, and an arrangement space for the pedal force **SW 81** etc. can be ensured without increasing the dimensions of the rotation lever **50**.

(2-1) Outline of Second Embodiment

[0056] In a second embodiment, configurations different from those of the first embodiment will be mainly described. In the second embodiment, the same elements as those of the first embodiment are denoted by the same reference numerals, and description of the same elements will not be repeated.

[0057] In the first embodiment described above, upon vehicle collision, the swaging pin **15** is broken to release the fixation between the rotation lever **50** and the second arm **40** via the small collision lever **60**. Instead, in the present embodiment, upon vehicle collision, the small collision lever **60** and the swaging pin **15** are disengaged from each other to release the fixation between the rotation lever **50** and the second arm **40** via the small collision lever **60**. In the present embodiment, the swaging pin **15** is an example of an engagement shaft portion.

[0058] As illustrated in FIG. 7, a cutout **63** to be engaged with the swaging pin **15** is formed in the lower end portion **60C** of the small collision lever **60**. The cutout **63** has a shape in which the edge of the lower end portion **60C** on the vehicle front side is cut out, and the inner diameter of the

cutout **63** is a size allowing penetration of the swaging pin **15**.

[0059] In the present embodiment, when the small collision lever **60** is at the restriction position, the swaging pin **15** at the offset portion **40A** of the second arm **40** is positioned on the vehicle rear side with respect to the second rotation shaft portion **14**. Specifically, the position of the swaging pin **15** at the offset portion **40A** is determined such that a straight line connecting the first rotation shaft portion **13** and the swaging pin **15** is perpendicular to a straight line connecting the second rotation shaft portion **14** and the swaging pin **15**.

[0060] When the small collision lever **60** is at the restriction position, the swaging pin **15** is engaged with the cutout **63**, so that the rotation lever **50** and the second arm **40** are fixed via the small collision lever **60**. On the other hand, when the collision bracket **200** contacts the second contact portion **60A** of the small collision lever **60**, the small collision lever **60** is at the release position, and the swaging pin **15** and the cutout **63** are disengaged from each other. At this time, immediately after the start of rotation of the small collision lever **60**, the cutout **63** moves between the vehicle rear side and the vehicle upper side along the straight line connecting the first rotation shaft portion **13** and the swaging pin **15**, so that the swaging pin **15** can be quickly removed from the cutout **63**. As a result, without breaking the swaging pin **15**, the rotation lever **50** and the second arm **40** fixed via the small collision lever **60** can be released, the rotation lever **50** can be rotated at the first rotation shaft portion **13**, and the step portion **21** can be displaced to the vehicle front side.

[0061] In the present embodiment described above, upon vehicle collision, the fixation between the rotation lever **50** and the second arm **40** via the small collision lever **60** can be released without breaking the swaging pin **15**. Thus, the shapes of the second arm **40** and the rotation lever **50** and the positions of the shaft portions can be designed without considering, e.g., the breaking load of the swaging pin **15**, and the degree of flexibility in design can be increased.

[0062] When the small collision lever **60** is at the restriction position, the position of the swaging pin **15** at the second arm **40** is determined such that the straight line connecting the first rotation shaft portion **13** and the swaging pin **15** is perpendicular to the straight line connecting the second rotation shaft portion **14** and the swaging pin **15**. With this configuration, immediately after the start of rotation of the small collision lever **60**, the cutout **63** moves between the vehicle rear side and the vehicle upper side along the straight line connecting the first rotation shaft portion **13** and the swaging pin **15**, so that the swaging pin **15** can be easily removed from the cutout **63** and operation failure at the time of prevention of backward movement can be reduced in the brake pedal device **1**.

(2-2) Modification of Second Embodiment

[0063] By providing a cutout in the second arm **40**, the fixation between the rotation lever **50** and the second arm **40** via the small collision lever **60** may be released upon vehicle collision. As illustrated in FIG. **8**, a cutout **46** to be engaged with the swaging pin **15** is formed in the offset portion **40A** of the second arm **40**. The cutout **46** has a shape in which the edge of the offset portion **40A** on the vehicle rear side is cut out, and the inner diameter of the cutout **46** is a size allowing penetration of the swaging pin **15**.

[0064] In the present embodiment, when the small collision lever **60** is at the restriction position, the swaging pin **15** penetrating through the small collision lever **60** is engaged with the cutout **46** of the second arm **40**, so that the rotation lever **50** and the second arm **40** are fixed via the small collision lever **60**. On the other hand, when the collision bracket **200** contacts the second contact portion **60A** of the small collision lever **60**, the lower end portion **60C** is rotated about the second rotation shaft portion **14** to the vehicle rear side, together with the swaging pin **15**. Also in the present embodiment, immediately after the start of rotation of the small collision lever **60**, the cutout **63** moves between the vehicle rear side and the vehicle upper side along the straight line connecting the first rotation shaft portion **13** and the swaging pin **15**, so that the swaging pin **15** can be quickly removed from the cutout **46**. With this configuration, the fixation between the rotation lever **50** and the second arm **40** via the small collision lever **60** is released, the rotation lever **50** is

rotated at the first rotation shaft portion **13**, and the step portion **21** is displaced to the vehicle front side.

(3) Other Embodiments

[0065] The technology disclosed in the present specification is not limited to the above-described embodiments, and can be modified into various forms without departing from the gist of the technology disclosed in the present specification and, for example, can be modified as follows.

[0066] In the above-described embodiments, the swaging pin **15** is used as the fixing member. Instead, the rotation lever **50** and the second arm **40** may be fixed to each other or released from each other by a bolt and a nut.

[0067] In the above-described embodiments, the brake pedal device **1** includes the pedal force detection device **80**. Instead, the brake pedal device **1** does not necessarily include the pedal force detection device **80**.

[0068] In the above-described embodiments, in the operation pedal mechanism **20**, the first arm **22** and the second arm **40** are coupled to each other by the link member **30**. Instead, the operation pedal mechanism **20** may include only the first arm. In this case, the first arm rotatably supports the rotation lever **50** at the first rotation shaft portion **13**, and the first arm is fixed to the rotation lever **50** via the swaging pin **15** at an upper end portion of the first arm.

[0069] In the above-described embodiments, the first arm **22** is a brake pedal, but may be a pedal (e.g., accelerator pedal or clutch pedal) used for vehicles.

[0070] The material of each of the components of the brake pedal device **1** is not limited to metal, and each of the components may be made of resin.

LIST OF REFERENCE SIGNS

[0071] **1** Brake Pedal Device [0072] **10** Pedal Bracket [0073] **11** Operation Shaft Portion [0074] **12** Intermediate Shaft Portion [0075] **15** Swaging Pin [0076] **20** Operation Pedal Mechanism [0077] **21** Step Portion [0078] **22** First Arm [0079] **30** Link Member [0080] **40** Second Arm [0081] **40A** Offset Portion [0082] **50** Rotation Lever [0083] **50A** First Contact Portion [0084] **51** Clevis Holding Member [0085] **54** Cutout [0086] **60** Small Collision Lever [0087] **60A** Second Contact Portion [0088] **70** Clevis [0089] **80** Pedal Force Detection Device [0090] **81** Pedal Force SW [0091] **82** Swing Lever [0092] **90** Operating Rod [0093] **200** Collision Bracket [0094] P Dash Panel

Claims

1. A vehicular operation pedal device, comprising: a support member fixed to a first vehicle component; an operation pedal mechanism provided on the support member and having a step portion that is rotatable relative to the support member and an operation lever that rotates toward a vehicle front side when the step portion is stepped on toward the vehicle front side; a rotation lever having a first contact portion extending to a vehicle rear side, supported on a vehicle lower side with respect to the first contact portion by a first rotation shaft portion so as to be rotatable toward the vehicle front side relative to the operation lever, and holding an operating rod protruding to the vehicle rear side from the first vehicle component; a differential lever having a second contact portion extending to the vehicle rear side and supported on the vehicle lower side with respect to the second contact portion so as to be rotatable toward the vehicle front side relative to the rotation lever by a second rotation shaft portion; and a fixing member that fixes the differential lever and the operation lever, wherein the differential lever is, when fixed to the operation lever by the fixing member, at a restriction position at which the second contact portion protrudes to the vehicle rear side beyond the first contact portion of the rotation lever, when the first vehicle component is displaced toward the vehicle rear side upon vehicle collision, a second vehicle component positioned on the vehicle rear side with respect to the first vehicle component contacts the second contact portion, so that the second contact portion is positioned on the vehicle front side with

respect to the first contact portion, and the differential lever is displaced to a release position at which a load corresponding to contact of the second vehicle component with the second contact portion is applied to the fixing member, and fixation between the differential lever and the operation lever by the fixing member is released, so that the second vehicle component contacts the first contact portion and the rotation lever rotates toward the vehicle front side at the first rotation shaft portion.

2. The vehicular operation pedal device according to claim 1, wherein the fixing member has an engagement shaft portion penetrating through the operation lever in a vehicle width direction, and a cutout provided in the differential lever and cut out on the vehicle front side, and the fixing member fixes the differential lever and the operation lever by engagement of the engagement shaft portion with the cutout.

3. The vehicular operation pedal device according to claim 1, wherein the fixing member has an engagement shaft portion penetrating through the differential lever in a vehicle width direction, and a cutout provided in the operation lever and cut out on the vehicle rear side, and the fixing member fixes the differential lever and the operation lever by engagement of the engagement shaft portion with the cutout.

4. The vehicular operation pedal device according to claim 1, further comprising: a coupling member fixed to a tip end of the operating rod; and a holding member holding the coupling member on the rotation lever, wherein the operation lever has a clearance hole, the holding member holds the coupling member on the rotation lever in a state of being housed in the clearance hole of the operation lever, and the first rotation shaft portion is positioned on a vehicle upper side with respect to a line segment in a direction in which the operating rod extends when the differential lever is at the restriction position and the step portion is at an initial position at which stepping operation is not performed on the step portion.

5. The vehicular operation pedal device according to claim 4, further comprising: a pedal force detection device having a swing lever that swings in response to reactive force from the operating rod and a pedal force switch that detects a pedal force applied to the step portion based on a swing amount of the swing lever, wherein the holding member holds the coupling member on the rotation lever such that the coupling member swings in a direction in which the reactive force from the operating rod is applied, and the swing lever is swingably held by the holding member together with the coupling member.

6. The vehicular operation pedal device according to claim 1, wherein the operation lever has, at an end portion on a vehicle upper side, an offset portion extending so as to be offset to the vehicle rear side with respect to the first rotation shaft portion, and the fixing member is positioned at the offset portion.

7. The vehicular operation pedal device according to claim 2, wherein a position of the engagement shaft portion is set such that a straight line connecting the first rotation shaft portion and the engagement shaft portion and a straight line connecting the second rotation shaft portion and the engagement shaft portion are perpendicular to each other when the differential lever is at the restriction position.
